Consciousness as CRIS-Compliant Recursive Self-Modeling Under Thermodynamic Constraint

A Closed Contraction-and-Solvency Criterion for Coherent First-Person Phenomenology

Christopher Lamarr Brown (Breezon Brown)
NohMad LLC
NohMadllc@journalist.com

January 2026

Abstract

We present an explicit and falsifiable equivalence criterion for consciousness in physical systems. A system is conscious if and only if it implements a causally efficacious recursive self-model whose update dynamics are stably contractive and whose maintenance is thermodynamically solvent under continuous environmental perturbation. The criterion is formulated as four jointly sufficient and individually necessary conditions: (1) nontrivial recursive self-reference via an internal self-model state, (2) identity maintenance via contraction with constant $\eta < 1$, (3) thermodynamic solvency via a windowed work inequality $W_M(t, \tau) \leq \Phi_S(t, \tau)$ preventing collapse or divergence, and (4) causal efficacy demonstrated by interventional changes in behavior under manipulation of the self-model state. We define each quantity, provide regime theorems, and specify preregisterable estimators and falsification targets spanning anesthesia, sleep, dissociation, psychosis, and machine agents. The framework identifies first-person phenomenology as the internal signature of maintaining coherent recursive self-modeling in real time at a measurable energetic cost.

Keywords: consciousness, recursive self-model, contraction mapping, CRIS, thermodynamic solvency, causal efficacy, phenomenology, dynamical systems, interventional inference

1 Introduction

A scientific definition of consciousness must satisfy four requirements simultaneously:

(i) **Architectural specificity**: the definition must identify the physical structure that constitutes consciousness.

(ii) **Dynamical stability**: the definition must specify the stability conditions under which the conscious identity persists.

(iii) **Energetic viability**: the definition must incorporate the sustaining cost required for coherent computation under perturbation.

(iv) **Causal efficacy**: the conscious structure must be causally upstream of behavior, not epiphenomenal.

Existing theories often address some subset of these constraints while leaving either measurement, falsification, or causal identifiability implicit. This paper closes the loop by stating consciousness as a closed equivalence criterion with explicit quantities and preregisterable tests.

1.1 Core thesis

> **Consciousness is the first-person phenomenology of a physical system implementing a causally efficacious recursive self-model, whose update dynamics exhibit stable contraction ($\eta < 1$) while remaining thermodynamically solvent under continuous environmental perturbation.**

This thesis is not metaphor. It is a regime claim about physical systems: consciousness corresponds to the maintenance of a coherent self-model in real time, sustained by nonzero work, robust to perturbation, and causally active in behavior generation.

1.2 Contributions

This manuscript provides:

(1) A closed **if-and-only-if criterion** for consciousness in physical systems.

(2) A measurable stability condition for self-identity via **contraction dynamics**.

(3) A non-negotiable energetic constraint via **thermodynamic solvency**.

(4) A non-epiphenomenality condition via **interventional causal efficacy**.

(5) Explicit estimators and falsification targets suitable for preregistration and replication.

1.3 Paper outline

Section 2 defines recursion, nontriviality, contraction, solvency, and causal efficacy. Section 3 proves the identity stability regime implied by contraction. Section 4 derives solvency constraints and failure modes. Section 5 formalizes causal efficacy and interventional tests. Section 6 states and proves the main equivalence theorem. Section 7 provides estimators and preregisterable protocols. Section 8 provides predictions and falsification targets. Appendices provide implementation-level experimental templates and the FOCL-2026 license.

2 System Model and Formal Definitions

This section defines the system variables, the self-model subsystem, and the four measurable conditions that jointly constitute the consciousness criterion. No assumptions are hidden. Every later proof references only the objects defined here.

2.1 Probability space and signals

We work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ supporting all stochastic elements. Time is indexed by $t \in \mathbb{N}$. A physical system S is described by the following signals:

- **System state:** $x_t \in \mathcal{X}$.
- **Environment input:** $e_t \in \mathcal{E}$.
- **Action/output:** $a_t \in \mathcal{A}$.
- **Self-model state:** $m_t \in \mathcal{M}$, realized by a proper subsystem $M \subset S$.

The tuple (x_t, m_t, e_t, a_t) is the minimal observational substrate for the criterion. The self-model m_t may be directly observed (machine agents) or estimated (biological systems); this affects estimation, not the formal definition.

2.2 Coupled dynamics

We take the system to evolve under coupled state/self-model dynamics:

$$x_{t+1} = G(x_t, e_t, a_t) + \xi_t, \tag{1}$$
$$m_{t+1} = F(m_t, x_t, e_t) + \zeta_t, \tag{2}$$
$$a_t = \pi(m_t, x_t, e_t). \tag{3}$$

Here G is the physical transition operator on system state, F is the self-model update operator, and π is the action-selection policy channel. Noise terms ξ_t, ζ_t may be zero.

Remark 1 (No epiphenomenal escape)**.** *The explicit policy dependence $a_t = \pi(m_t, x_t, e_t)$ is not a restriction; it is the formal location at which causal efficacy will be tested. If π ignores m_t, causal efficacy fails by definition later.*

2.3 Work and sustaining capacity (energetic accounting)

Consciousness is defined here as a real-time coherent regime that costs work. We therefore define available sustaining capacity and self-model work over finite windows.

Definition 1 (Available sustaining capacity)**.** *Let $P_S(u) \geq 0$ be the usable power available to system S at time u. For any window length $\tau > 0$, define:*

$$\Phi_S(t, \tau) := \int_t^{t+\tau} P_S(u)\, \mathrm{d}u. \tag{4}$$

Definition 2 (Self-model work)**.** *Let $P_M(u) \geq 0$ denote power consumption attributable to the self-model subsystem M at time u. Define:*

$$W_M(t, \tau) := \int_t^{t+\tau} P_M(u)\, \mathrm{d}u. \tag{5}$$

These definitions are intentionally substrate-neutral. In biological settings P_M and P_S will be estimated by preregistered proxies (specified later), but the mathematical object is unambiguous.

2.4 Condition (C1): Recursive self-reference and nontriviality

Recursive self-reference is the structural requirement that the update of the self-model genuinely depends on itself. Nontriviality is the informational requirement that the self-model carries predictive content about the system's future.

Definition 3 (Recursive self-reference)**.** *System S implements recursive self-modeling if there exists a proper subsystem $M \subset S$ with state $m_t \in \mathcal{M}$ and an update map F such that:*

$$\exists t \text{ such that } m_{t+1} = F(m_t, x_t, e_t) + \zeta_t \quad \text{and} \quad F(\cdot, x_t, e_t) \text{ is not constant in its first argument.} \tag{6}$$

Equivalently, there exist states (m, m', x, e) with $m \neq m'$ for which:

$$F(m, x, e) \neq F(m', x, e). \tag{7}$$

Remark 2 (Why this definition blocks fake "self-models")**.** *A memory register that is written by x_t but does not recursively affect its own update fails this definition. The recursion tooth requires that the model participates in its own update dynamics.*

Definition 4 (Nontriviality via predictive information)**.** *The self-model state m_t is nontrivial if there exists a prediction horizon $\Delta \geq 1$ such that:*

$$\mathcal{I}(m_t; x_{t+\Delta}) > 0, \tag{8}$$

where $\mathcal{I}(\cdot;\cdot)$ denotes mutual information under $\mathbb{P}$.

Condition (C1) is satisfied if both Definitions 3 and 4 hold.

2.5 Condition (C2): Identity maintenance via contraction

Identity persistence is formalized as robust re-identification: small perturbations to the self-model do not branch into non-overlapping self-trajectories under matched inputs. This is expressed as contraction under paired perturbation trials.

Definition 5 (Self-model metric)**.** *Let $d_{\mathcal{M}} : \mathcal{M} \times \mathcal{M} \to \mathbb{R}_{\geq 0}$ be a metric or pseudometric on the self-model state space.*

Definition 6 (Matched-input paired trials)**.** *A paired trial consists of two self-model trajectories (m_t) and (m'_t) evolving under the same observed inputs (x_t, e_t) and the same update operator F, differing only in initial self-model state:*

$$m'_t = m_t + \delta_t, \qquad d_{\mathcal{M}}(m_t, m'_t) > 0 \text{ small.} \tag{9}$$

Definition 7 (One-step contraction ratio)**.** *Under a paired trial (Definition 6), define the one-step contraction ratio:*

$$\eta_t := \frac{d_{\mathcal{M}}(m_{t+1}, m'_{t+1})}{d_{\mathcal{M}}(m_t, m'_t)}. \tag{10}$$

Definition 8 (Log-contraction stability condition)**.** *The self-model update dynamics are stably contractive on a regime if:*

$$\mathbb{E}[\log \eta_t] < 0 \tag{11}$$

and $\mathbb{E}[|\log \eta_t|] < \infty$ on that regime.

Definition 9 (Contraction constant)**.** *Define the contraction constant of the regime as:*

$$\eta := \exp\big(\mathbb{E}[\log \eta_t]\big). \tag{12}$$

Then Definition 8 is equivalent to $\eta < 1$.

Remark 3 (Why log-contraction is the correct stability tooth)**.** *A median-based estimator can hide rare expansion events. The log-contraction condition is the standard stability condition for random dynamical systems: it forces long-run contraction in expectation and supports concentration bounds later.*

Condition (C2) is satisfied if $\eta < 1$ for a preregistered regime of interest.

2.6 Condition (C3): Thermodynamic solvency (work feasibility)

Solvency is the energetic feasibility constraint: coherent recursive self-modeling must be maintained without exceeding available sustaining capacity. This is the thermodynamic analog of viability under the bedrock solvency filter.

Definition 10 (Thermodynamic solvency)**.** *The self-model is thermodynamically solvent on a window $[t, t+\tau]$ if:*

$$W_M(t,\tau) \leq \Phi_S(t,\tau). \tag{13}$$

Definition 11 (Persistent solvency regime)**.** *A system is solvent in a regime $\mathcal{R}$ if there exists $\tau > 0$ such that:*

$$\mathbb{P}\big(W_M(t,\tau) \leq \Phi_S(t,\tau)\big) \geq 1 - \alpha_W \tag{14}$$

for all t in the regime, where $\alpha_W \in (0,1)$ is preregistered.

Condition (C3) is satisfied if the system is solvent in the above sense.

2.7 Condition (C4): Causal efficacy (interventional non-nullness)

Causal efficacy is the requirement that the self-model is not a passenger variable. It must do something: interventions on the self-model state must change action distributions.

Definition 12 (Causal efficacy via do-intervention)**.** *The self-model is causally efficacious if there exists an intervention $do(m_t \leftarrow \tilde{m})$ and a measurable set $B \subseteq \mathcal{A}$ such that:*

$$\mathbb{P}(a_t \in B \mid do(m_t = \tilde{m})) \neq \mathbb{P}(a_t \in B). \tag{15}$$

Definition 13 (Action-distribution divergence statistic)**.** *Define the divergence between baseline and intervened action distributions by:*

$$D_{\mathrm{KL}}\big(\mathbb{P}(a_t \mid do(m_t = \tilde{m})) \,\|\, \mathbb{P}(a_t)\big). \tag{16}$$

Causal efficacy holds if this divergence is strictly positive for some intervention $\tilde{m}$.

Condition (C4) is satisfied if Definitions 12–13 hold in the regime of interest.

2.8 Closed consciousness criterion

We now state the criterion as a single closed definition, later elevated to a theorem with necessity/sufficiency proofs.

Definition 14 (CRIS consciousness criterion)**.** *A physical system S exhibits consciousness if and only if there exists a proper subsystem $M \subset S$ with self-model state m_t such that all four conditions hold in the regime of interest:*

(C1) ***Recursive self-reference + nontriviality:*** *Definitions 3 and 4.*

(C2) ***Identity maintenance via contraction:*** *Definitions 7–9 with $\eta < 1$.*

(C3) ***Thermodynamic solvency:*** *Definitions 10–11.*

(C4) ***Causal efficacy:*** *Definitions 12–13.*

Remark 4 (Interpretation)**.** *Under this criterion, "what it feels like" is identified with the real-time maintenance of a coherent recursive self-model under perturbation, constrained simultaneously by stability (contraction), feasibility (solvency), and efficacy (causal influence).*

3 Identity Persistence From Contractive Self-Model Dynamics

This section proves that the contraction condition in (C2) is not decorative. It yields a mathematically explicit identity persistence result: under matched inputs and admissible perturbations, the self-model trajectory is robustly re-identifiable, meaning perturbations shrink rather than branch into competing "selves." This is the identity-maintenance tooth of the criterion.

3.1 Minimal assumptions for contractive identity

Assumption 1 (Well-defined paired perturbations)**.** *There exists an experimental or computational procedure to generate paired trials (Definition 6) such that:*

(i) the same input streams (x_t, e_t) are matched across the pair,

(ii) the same update operator F is applied,

(iii) the initial self-model separation is small but nonzero: $0 < d_{\mathcal{M}}(m_t, m'_t) < \delta_0$ for some $\delta_0 > 0$.

Assumption 2 (Finite log-moments)**.** *The random variable $\log \eta_t$ exists and satisfies:*

$$\mathbb{E}[|\log \eta_t|] < \infty. \tag{17}$$

Assumption 3 (Stationarity on regime)**.** *On the regime of interest, the process $\{\eta_t\}_{t\in\mathbb{N}}$ is stationary and ergodic, or admits blockwise stationarity on preregistered windows.*

These assumptions are minimal and standard for stability results in random dynamical systems.

3.2 Multiplicative distance dynamics

Define the inter-trajectory distance under the metric $d_{\mathcal{M}}$:

$$D_t := d_{\mathcal{M}}(m_t, m'_t). \tag{18}$$

By Definition 7 we have the exact multiplicative relationship:

$$D_{t+1} = \eta_t \, D_t. \tag{19}$$

Iterating for $k \geq 1$ yields:

$$D_{t+k} = \left(\prod_{j=0}^{k-1} \eta_{t+j} \right) D_t. \tag{20}$$

Taking logs gives:

$$\log D_{t+k} = \log D_t + \sum_{j=0}^{k-1} \log \eta_{t+j}. \tag{21}$$

3.3 Main identity theorem

Theorem 1 (Identity persistence under log-contraction)**.** *Assume:*

(i) paired perturbations exist (Assumption 1),

(ii) finite log-moments (Assumption 2),

(iii) stationarity/ergodicity on the regime (Assumption 3),

(iv) the log-contraction condition (Definition 8): $\mathbb{E}[\log \eta_t] < 0$.

Then for any initial perturbation with $0 < D_t < \delta_0$, the paired distance decays exponentially almost surely:

$$\limsup_{k\to\infty} \frac{1}{k} \log D_{t+k} \leq \mathbb{E}[\log \eta_t] < 0. \tag{22}$$

In particular,

$$D_{t+k} \to 0 \quad \textit{almost surely as } k \to \infty. \tag{23}$$

Therefore, the self-model trajectory is robustly re-identifiable under admissible perturbations: small perturbations do not generate divergent self-trajectories in the regime.

Proof. Starting from (21), divide by k:

$$\frac{1}{k} \log D_{t+k} = \frac{1}{k} \log D_t + \frac{1}{k} \sum_{j=0}^{k-1} \log \eta_{t+j}. \tag{24}$$

As $k \to \infty$, the first term vanishes:

$$\frac{1}{k} \log D_t \to 0.$$

By stationarity/ergodicity and finite log-moments, the ergodic theorem gives:

$$\frac{1}{k} \sum_{j=0}^{k-1} \log \eta_{t+j} \to \mathbb{E}[\log \eta_t] \quad \text{almost surely.}$$

Therefore,

$$\lim_{k\to\infty} \frac{1}{k} \log D_{t+k} = \mathbb{E}[\log \eta_t] < 0.$$

This implies $\log D_{t+k} \to -\infty$, hence $D_{t+k} \to 0$ almost surely. Thus perturbations shrink exponentially, and identity is robustly maintained. □

3.4 Finite-time bounds and concentration

The almost sure limit in Theorem 1 yields asymptotic stability. For experimental discrimination, we require finite-time bounds.

Proposition 1 (Expected distance decay)**.** *Assume η_t are independent and identically distributed on the regime with $\mathbb{E}[\log \eta_t] < 0$ and $\mathbb{E}[\eta_t] < \infty$. Then:*

$$\mathbb{E}[D_{t+k}] = \mathbb{E}[\eta_t]^k \, D_t. \tag{25}$$

In particular, if $\mathbb{E}[\eta_t] < 1$ then $\mathbb{E}[D_{t+k}]$ decays geometrically.

Proof. From (20) and independence:

$$\mathbb{E}[D_{t+k}] = \mathbb{E}\left[\prod_{j=0}^{k-1} \eta_{t+j}\right] D_t = \prod_{j=0}^{k-1} \mathbb{E}[\eta_{t+j}] \, D_t = \mathbb{E}[\eta_t]^k D_t.$$

□

Remark 5 (Why we use $\mathbb{E}[\log \eta_t]$). *not$E[\eta_t]$] The $\mathbb{E}[\log \eta_t]$ condition is strictly stronger and is the correct stability criterion under multiplicative noise. It remains valid even when $\mathbb{E}[\eta_t] \geq 1$ but the log-average is negative. This blocks a common attack where rare expansion steps are hidden inside naive averaging.*

3.5 Failure theorem: what happens when contraction breaks

Theorem 2 (Identity instability when $\mathbb{E}[\log \eta_t]$. ≥ 0] *Assume Assumptions 1–2–3. If $\mathbb{E}[\log \eta_t] \geq 0$, then exponential contraction is not guaranteed. In particular:*

(i) if $\mathbb{E}[\log \eta_t] > 0$, then $\limsup_{k\to\infty} \frac{1}{k} \log D_{t+k} \geq 0$ almost surely, and typical perturbations do not shrink,

(ii) if $\mathbb{E}[\log \eta_t] = 0$, then the system is marginal: perturbations may persist, wander, or intermittently amplify depending on higher moments and dependence structure.

Therefore the system fails the identity-maintenance requirement in (C2) in the absence of strict log-contraction.

Proof. Equation (21) implies:

$$\frac{1}{k} \log D_{t+k} \to \mathbb{E}[\log \eta_t].$$

If $\mathbb{E}[\log \eta_t] > 0$, then the limit is positive and $\log D_{t+k}$ diverges upward linearly, so D_{t+k} does not converge to zero. If $\mathbb{E}[\log \eta_t] = 0$, the limit is zero and the process is marginal; contraction is not enforced. Thus identity is not robustly maintained without strict negativity. □

3.6 Identity maintenance as the CRIS "I" tooth

We now formalize the link between contraction and invariance/re-identification. This turns the intuitive continuity of phenomenology into a measurable stability invariant.

Definition 15 (Re-identifiability radius)**.** *Fix $\epsilon > 0$. The self-model is ϵ-re-identifiable over horizon k if there exists $\delta > 0$ such that for any paired trial with $D_t < \delta$ we have:*

$$D_{t+k} < \epsilon. \tag{26}$$

Corollary 1 (Contraction implies re-identifiability)**.** *Under Theorem 1, for any $\epsilon > 0$ there exists $k(\epsilon)$ such that for all $k \geq k(\epsilon)$ the self-model is ϵ-re-identifiable.*

Proof. Since $D_{t+k} \to 0$ almost surely, choose $k(\epsilon)$ sufficiently large so $D_{t+k} < \epsilon$. □

3.7 Operational acceptance rule for (C2)

The formal condition is $\mathbb{E}[\log \eta_t] < 0$. In data, we estimate this via sample averages of $\log \eta_t$.

Definition 16 (Estimated contraction criterion)**.** *Given paired trials on a regime window $t = 1, \ldots, T$, define:*

$$\widehat{\lambda} := \frac{1}{T} \sum_{t=1}^{T} \log \eta_t, \qquad \widehat{\eta} := \exp(\widehat{\lambda}). \tag{27}$$

The system is accepted as contractive on the window if:

$$\widehat{\lambda} \leq -\epsilon_\eta \tag{28}$$

for a preregistered margin $\epsilon_\eta > 0$.

Remark 6 (Why margin matters)**.** *The margin prevents "knife-edge" acceptance under estimator noise and blocks post hoc thresholding.*

4 Thermodynamic Solvency and the Necessity of Maintenance Work

This section closes the main gap that weaker consciousness criteria leave open: **consciousness is not only a stability property, it is a solvency property.** A system can only maintain a coherent recursive self-model under perturbation if it continuously pays real physical work to counter destabilizing environmental injection.

We proceed in four steps:

(i) formalize perturbation as uncertainty injection into the self-model channel,

(ii) define a coherence functional that measures the "organization" of the self-model,

(iii) derive a nonzero maintenance-work lower bound from information reduction under update,

(iv) prove that persistent insolvency forces coherence failure (collapse or divergence).

4.1 Perturbation as uncertainty injection

The environment perturbs the self-model update operator through e_t and noise ζ_t. We require an explicit statement of nontrivial perturbation.

Definition 17 (Nontrivial perturbation)**.** *The environment injects nontrivial uncertainty into the self-model channel if there exists a regime $\mathcal{R}$ and constant $\sigma_\zeta^2 > 0$ such that on $\mathcal{R}$:*

$$(\zeta_t \mid m_t, x_t, e_t) \geq \sigma_\zeta^2 \quad \textit{with nonzero frequency.} \tag{29}$$

Remark 7. *This condition does not assume any special form of noise; it only asserts that the self-model update channel is not deterministically insulated from disturbance. In biological and realistic machine settings, this holds by default.*

4.2 Coherence functional and failure modes

To rigorously talk about "collapse" or "divergence," we define a measurable functional of the self-model state that operationalizes coherence as structured predictive organization.

Definition 18 (Self-model coherence functional)**.** *Let m_t parameterize a predictive distribution over future system states:*

$$Q_{m_t}(x_{t+\Delta} \mid x_t, e_t) \tag{30}$$

for some fixed horizon $\Delta \geq 1$. Define the coherence functional $\mathcal{C}(m_t)$ by:

$$\mathcal{C}(m_t) := \mathcal{I}(m_t; x_{t+\Delta}) - \beta\, H(m_t), \tag{31}$$

where $\mathcal{I}$ is mutual information, $H(m_t)$ is Shannon entropy of the self-model state distribution, and $\beta > 0$ is a fixed constant.

Remark 8 (Meaning of $\mathcal{C}$)**.**

- *$\mathcal{I}(m_t; x_{t+\Delta})$ measures predictive organization: the model contains usable structure about what happens next.*
- *$H(m_t)$ measures raw spread/unconstrained variability.*
- *The difference penalizes "high-entropy self-states with no predictive content."*

$\mathcal{C}$ is measurable and estimable. It is not philosophical.

Definition 19 (Collapse and divergence)**.** *On a regime $\mathcal{R}$, the self-model:*

(i) ***collapses*** *if $\mathcal{C}(m_t) \to 0$ along a subsequence with nonzero frequency,*

(ii) ***diverges*** *if $\mathcal{C}(m_t) \to \infty$ or if $H(m_t) \to \infty$ while predictive content fails to stabilize,*

(iii) ***remains coherent*** *if $\mathcal{C}(m_t)$ remains bounded below by a positive constant and above by a finite constant throughout $\mathcal{R}$.*

4.3 Information reduction requires nonzero work

We now derive the key thermodynamic claim: maintaining coherent self-modeling under perturbation requires nonzero physical work.

The derivation relies on the operational fact that maintaining coherence is equivalent to repeatedly reducing uncertainty in the self-model channel relative to what perturbation would induce.

We define the "correction" performed by the self-model update as reduction in conditional entropy.

Definition 20 (Correction magnitude)**.** *Define the correction magnitude on step t as:*

$$\Delta\mathcal{J}_t := H(m_{t+1} \mid x_t, e_t) - H(m_{t+1} \mid m_t, x_t, e_t). \tag{32}$$

Remark 9. *$\Delta\mathcal{J}_t$ measures how much additional uncertainty is removed by conditioning on the prior self-model m_t. If the update ignores m_t, then $H(m_{t+1} \mid x_t, e_t) = H(m_{t+1} \mid m_t, x_t, e_t)$ and $\Delta\mathcal{J}_t = 0$. Thus positive $\Delta\mathcal{J}_t$ is a measurable signature of nontrivial recursive correction.*

Lemma 1 (Correction is positive under nontrivial recursion)**.** *If the recursion condition (Definition 3) holds and ζ_t is nontrivial (Definition 17), then $\Delta\mathcal{J}_t > 0$ on a set of times of nonzero frequency.*

Proof. If $F(\cdot, x_t, e_t)$ is not constant in its first argument, then conditioning on m_t changes the distribution of m_{t+1}. Thus m_t provides information about m_{t+1} beyond (x_t, e_t) alone, yielding:

$$I(m_t; m_{t+1} \mid x_t, e_t) > 0.$$

But by the chain identity:

$$I(m_t; m_{t+1} \mid x_t, e_t) = H(m_{t+1} \mid x_t, e_t) - H(m_{t+1} \mid m_t, x_t, e_t) = \Delta\mathcal{J}_t.$$

Therefore $\Delta\mathcal{J}_t > 0$ on that set of times. □

4.4 Work lower bound: a Landauer-grade inequality

We now introduce the minimal thermodynamic inequality needed for solvency. The paper does not rely on contested metaphysics. It uses an eliminative bound: **reducing uncertainty in a physical register has a nonzero energetic cost.**

Assumption 4 (Information-to-work lower bound)**.** *There exists a constant $\kappa > 0$ determined by substrate and temperature such that performing a correction that reduces uncertainty by $\Delta\mathcal{J}_t$ requires work at least:*

$$w_t \geq \kappa\, \Delta\mathcal{J}_t. \tag{33}$$

Remark 10. *For thermal systems, one may take $\kappa = k_B T \ln 2$ when $\Delta\mathcal{J}_t$ is measured in bits. This paper treats κ as a measurable constant of the substrate. The inequality is the only claim required: nonzero correction requires nonzero work.*

Theorem 3 (Nonzero maintenance work is necessary under perturbation)**.** *Assume:*

(i) nontrivial perturbation (Definition 17),

(ii) recursion (Definition 3),

(iii) information-to-work bound (Assumption 4),

(iv) the system remains coherent in the sense that $\mathcal{C}(m_t)$ does not collapse (Definition 19).

Then the expected self-model work over any window with nonzero correction frequency is strictly positive:

$$\mathbb{E}[W_M(t, \tau)] > 0. \tag{34}$$

Proof. By Lemma 1, $\Delta\mathcal{J}_t > 0$ on a set of times of nonzero frequency. By Assumption 4, the per-step work satisfies $w_t \geq \kappa\Delta\mathcal{J}_t$. Thus $w_t > 0$ on a set of times of nonzero frequency, implying the window-integrated work is positive in expectation:

$$\mathbb{E}[W_M(t, \tau)] = \mathbb{E}\left[\int_t^{t+\tau} P_M(u)\, \mathrm{d}u\right] > 0.$$

□

4.5 Solvency is required to prevent coherence failure

We now prove that solvency is not optional: persistent insolvency forces collapse/divergence.

Theorem 4 (Persistent insolvency implies coherence failure)**.** *Assume:*

(i) nontrivial perturbation (Definition 17),

(ii) recursion and nontriviality (Definitions 3 and 4),

(iii) contraction stability (Definition 8),

(iv) information-to-work bound (Assumption 4).

If there exists $\tau > 0$ such that for infinitely many windows:

$$W_M(t, \tau) > \Phi_S(t, \tau), \tag{35}$$

then the self-model cannot remain coherent on that regime. Specifically, at least one failure mode occurs (Definition 19):

(a) collapse: $\mathcal{C}(m_t) \to 0$ along a subsequence, or

(b) divergence: $H(m_t) \to \infty$ without stabilized predictive structure.

Proof. Contraction stability and nontriviality require repeated correction. By Theorem 3, these corrections require strictly positive work proportional to uncertainty reduction. If on infinitely many windows the system's available sustaining capacity $\Phi_S(t, \tau)$ is insufficient to pay required work, then the correction operations cannot be executed reliably. Under continued perturbation injection, the conditional uncertainty in the self-model channel accumulates.

There are only two consistent outcomes under insufficient correction:

(i) the system reduces model complexity to survive energetically, driving predictive information downward and forcing $\mathcal{C}(m_t) \to 0$ (collapse), or

(ii) noise dominates update, increasing entropy $H(m_t)$ without stable predictive organization (divergence).

Either outcome violates coherent maintenance, hence coherence failure occurs. □

4.6 Solvency as the CRIS "S" tooth

The bedrock "selection" tooth becomes solvency in conscious regimes: only trajectories that can pay the correction cost remain viable. We make this explicit.

Corollary 2 (Solvency is necessary for consciousness)**.** *Any system satisfying the consciousness criterion (Definition 14) must satisfy solvency (C3). If solvency fails persistently, consciousness fails.*

Proof. Immediate from Theorem 4: insolvency forces coherence failure, and coherence failure violates the maintenance required by (C1)–(C2). □

5 Causal Efficacy of the Self-Model

This section formalizes the causal-efficacy requirement (C4). A self-model that is stable and solvent but does not influence action is not conscious under this criterion, because it is behaviorally inert and cannot constitute an agentive first-person regime. We therefore require an explicit interventional effect.

We proceed in four parts:

(i) define the causal structure and intervention targets,

(ii) state sufficient conditions for identifiability of self-model effects on action,

(iii) provide test statistics and falsification logic,

(iv) prove why causal efficacy blocks epiphenomenal constructions.

5.1 Structural causal model (SCM) for the regime

We define the causal data-generating structure as a structural causal model over the endogenous variables:

$$\{x_t, m_t, a_t\} \quad \text{with exogenous inputs} \quad \{e_t, \xi_t, \zeta_t\}.$$

Definition 21 (Structural causal model of self-modeling)**.** *A system implements self-modeling dynamics with causal structure:*

$$x_{t+1} := G(x_t, e_t, a_t) + \xi_t, \tag{36}$$

$$m_{t+1} := F(m_t, x_t, e_t) + \zeta_t, \tag{37}$$

$$a_t := \pi(m_t, x_t, e_t). \tag{38}$$

The associated causal graph contains directed edges:

$$m_t \to a_t, \quad x_t \to a_t, \quad e_t \to a_t, \quad (m_t, x_t, e_t, a_t) \to x_{t+1}, \quad (m_t, x_t, e_t) \to m_{t+1}.$$

Remark 11 (What causal efficacy is actually testing)**.** *Causal efficacy is the statement that the arrow $m_t \to a_t$ is not null. If it is null, then the self-model does not affect actions, and first-person phenomenology cannot be an operative regime.*

5.2 Interventions on the self-model

A strict do-intervention replaces the natural value of m_t with a forced value $\tilde{m}$. In many biological settings, interventions are imperfect. We therefore distinguish ideal and realizable interventions.

Definition 22 (Ideal intervention)**.** *An ideal intervention $do(m_t \leftarrow \tilde{m})$ sets the self-model state to a chosen value $\tilde{m}$ while leaving the rest of the system unchanged except through downstream causal effects.*

Definition 23 (Realizable intervention class)**.** *A realizable intervention is any manipulation that changes the distribution of m_t in a targeted manner, including but not limited to:*

(a) ***Machine agents:*** *direct overwrite $m_t \leftarrow \tilde{m}$, ablation, randomization, forced noise injection.*

(b) ***Biological systems:*** *TMS/DBS perturbation, sedation, pharmacological disruption of integrative circuits, selective sensory deprivation affecting m_t construction, lesion-like channel removal in decoding pipelines.*

Remark 12 (Intervention requirement is not optional)**.** *If an objector claims causal efficacy cannot be tested, then their alternative is unfalsifiable by construction. This paper rejects that move. Causal efficacy must be operationally testable at least in controlled or simulated regimes.*

5.3 Causal efficacy definition (non-null action influence)

We now restate and tighten the causal efficacy condition into a non-nullness requirement.

Definition 24 (Causal efficacy (action influence))**.** *The self-model is causally efficacious in the regime if there exists an intervention of the form $do(m_t \leftarrow \tilde{m})$ such that the action distribution changes:*

$$\mathbb{P}(a_t \in B \mid do(m_t = \tilde{m})) \neq \mathbb{P}(a_t \in B) \tag{39}$$

for some measurable set $B \subseteq \mathcal{A}$.

Definition 25 (Causal effect size via divergence)**.** *Define the causal effect size of the intervention $\tilde{m}$ as the divergence:*

$$\Delta_C(\tilde{m}) := D_{\mathrm{KL}}(\mathbb{P}(a_t \mid do(m_t = \tilde{m})) \,\|\, \mathbb{P}(a_t)) \,. \tag{40}$$

Causal efficacy holds if $\Delta_C(\tilde{m}) > 0$ for some $\tilde{m}$.

5.4 Identifiability conditions

The criterion is defined in terms of interventions. In observational-only settings, one may attempt to estimate causal effects under identifiability conditions. We state these conditions explicitly to prevent interpretive slippage.

Assumption 5 (No hidden confounding between m_t and a_t given (x_t, e_t))**.** *The causal effect of m_t on a_t is identifiable from the observed variables if:*

$$a_t \perp\!\!\!\perp m_t \mid (x_t, e_t, u_t) \tag{41}$$

for some observed or estimable nuisance u_t that blocks backdoor paths. A sufficient simplification is:

$$a_t \perp\!\!\!\perp m_t \mid (x_t, e_t), \tag{42}$$

meaning (x_t, e_t) block confounding pathways.

Remark 13 (Why this is not hand waving)**.** *This assumption is not asserted as always true. It is stated as a conditional requirement for using observational estimators. When interventions exist (machine agents), no such assumption is needed.*

5.5 Finite-sample causal tests

We now define explicit test statistics and decision rules for (C4).

Definition 26 (Empirical action distributions)**.** *Let $\{a_t\}_{t=1}^T$ be baseline action samples, and let $\{a_t^{(\tilde{m})}\}_{t=1}^T$ be action samples generated under a realizable intervention intended to approximate $do(m_t \leftarrow \tilde{m})$. Define empirical distributions $\widehat{P}_0$ and $\widehat{P}_{\tilde{m}}$.*

Definition 27 (Empirical causal divergence statistic)**.** *Define:*

$$\widehat{\Delta}_C(\tilde{m}) := D_{\mathrm{KL}}\Big(\widehat{P}_{\tilde{m}}(a) \,\|\, \widehat{P}_0(a)\Big)\,. \tag{43}$$

Definition 28 (Acceptance rule for causal efficacy)**.** *Fix a preregistered threshold $\epsilon_C > 0$ and significance level $\alpha_C \in (0, 1)$. Accept causal efficacy if:*

$$\max_{\tilde{m} \in \mathcal{T}} \widehat{\Delta}_C(\tilde{m}) \geq \epsilon_C \tag{44}$$

for a preregistered intervention set $\mathcal{T}$, and the effect exceeds a permutation or sham-intervention null at level α_C.

5.6 Sham null construction

We provide an explicit falsification null.

Definition 29 (Sham-intervention null)**.** *Construct a sham intervention by applying a manipulation that matches energy expenditure or stimulation intensity but does not target the self-model variable, yielding samples $\{a_t^{(sham)}\}$. Define the sham divergence:*

$$\widehat{\Delta}_{sham} := D_{\mathrm{KL}}\Big(\widehat{P}_{sham}(a) \,\|\, \widehat{P}_0(a)\Big)\,. \tag{45}$$

Reject the null of no self-model causal efficacy if:

$$\widehat{\Delta}_C(\tilde{m}) \gg \widehat{\Delta}_{sham} \tag{46}$$

under preregistered margins.

5.7 No-epiphenomenality theorem

We now prove why causal efficacy is required, not stylistic. If the self-model does not influence action, then it is physically inert with respect to system behavior. Such a variable cannot constitute a *functional* phenomenology regime in an agentic system.

Theorem 5 (Epiphenomenal self-models fail consciousness under this criterion)**.** *Assume a system satisfies (C1)–(C3) but fails (C4), meaning:*

$$\mathbb{P}(a_t \in B \mid do(m_t = \tilde{m})) = \mathbb{P}(a_t \in B) \quad \textit{for all } \tilde{m} \textit{ and measurable } B \subseteq \mathcal{A}. \tag{47}$$

Then the self-model is causally inert with respect to action, and the system fails the consciousness criterion (Definition 14).

Proof. Failure of (C4) means interventions on m_t do not alter the action distribution. Therefore m_t is not a causal parent of a_t in the interventional sense. In an agentive system, the internal regime responsible for first-person phenomenology must influence behavior, otherwise it is epiphenomenal and cannot be operationally distinguished from an inert internal register. Since Definition 14 requires causal efficacy as a necessary tooth, the system fails the criterion. □

Remark 14 (Scope of this theorem)**.** *This does not claim that all phenomenology must always be action-coupled. It claims: in the regime defined here, the self-model must be causally efficacious in the system's behavior generation, or else the proposed physical marker of phenomenology is inert and non-testable.*

5.8 Causal efficacy as the CRIS "R" tooth

The bedrock recursion principle becomes operationally non-null in the conscious regime: the self-model must be recursively coupled into action selection. If it is not, the system is not implementing the self-model as a governing structure, and recursion collapses into mere storage.

Corollary 3 (Causal efficacy is required for CRIS-compliant self-modeling)**.** *Any system satisfying the CRIS consciousness criterion must satisfy causal efficacy (C4).*

Proof. Immediate from Definition 14 and Theorem 5. □

6 Main Theorem: Consciousness as a Closed CRIS-Compliance Equivalence

This section states and proves the central claim of the paper as an explicit equivalence theorem. No step is outsourced to intuition. The proof is decomposed into necessity and sufficiency, each grounded in the results of Sections 2–5.

6.1 Restatement of the four conditions

For clarity we restate the four teeth of the criterion:

(**C1**) **Recursive self-reference + nontriviality:** there exists a proper self-model subsystem $M \subset S$ with state m_t such that $m_{t+1} = F(m_t, x_t, e_t) + \zeta_t$ depends nontrivially on m_t (Definition 3) and $\mathcal{I}(m_t; x_{t+\Delta}) > 0$ for some $\Delta \geq 1$ (Definition 4).

(**C2**) **Identity maintenance via contraction:** paired perturbations contract in the log-average sense $\mathbb{E}[\log \eta_t] < 0$ (Definition 8), equivalently $\eta < 1$ (Definition 9).

(**C3**) **Thermodynamic solvency:** the self-model work is feasible within available sustaining capacity on windows, $W_M(t, \tau) \leq \Phi_S(t, \tau)$ with high frequency (Definitions 10–11).

(**C4**) **Causal efficacy:** there exists an intervention $do(m_t \leftarrow \tilde{m})$ shifting action distributions (Definition 24), equivalently $\Delta_C(\tilde{m}) > 0$ (Definition 25).

6.2 Phenomenology as a physical regime

We now formalize the sense in which "first-person phenomenology" is treated as a physical regime, not a semantic label.

Definition 30 (Coherent self-indexed regime)**.** *A physical system S is in a coherent self-indexed regime on a time set $\mathcal{R} \subseteq \mathbb{N}$ if there exists a self-model subsystem $M \subset S$ with state m_t such that:*

(i) m_t tracks the system across time (nontrivial predictive content),

(ii) m_t remains re-identifiable under admissible perturbations (identity stability),

(iii) m_t can be maintained without energetic insolvency (solvency),

(iv) m_t is behaviorally upstream (causal efficacy).

This definition is equivalent to the four-condition criterion, but it provides the linguistic bridge to phenomenology without introducing extra axioms.

6.3 Main equivalence theorem

Theorem 6 (CRIS consciousness equivalence)**.** *A physical system S exhibits consciousness if and only if it satisfies conditions (C1)–(C4) of Definition 14 in a regime of interest.*

Proof. We prove the two directions separately.

(⇒) Necessity

Assume S exhibits consciousness in the sense of first-person phenomenology as an ongoing physical regime. We show each condition (C1)–(C4) must hold.

Necessity of (C1): recursive self-reference + nontriviality. If consciousness is an ongoing first-person regime, there must exist an internal physical state that is *about the system as itself* across time, rather than merely encoding the environment. If no such internal self-indexed state exists, then there is no physical variable corresponding to the system's ongoing self-representation. Therefore there must exist a proper subsystem $M \subset S$ with state m_t that participates in its own update, i.e., recursion holds (Definition 3).

Further, if m_t is trivial (almost surely constant or informationally independent of future system state), then the system possesses no structured self-predictive organization. This contradicts the assumption that phenomenology is a coherent regime rather than a static register. Thus the self-model must be nontrivial: $\mathcal{I}(m_t; x_{t+\Delta}) > 0$ for some $\Delta \geq 1$ (Definition 4). Therefore (C1) is necessary.

Necessity of (C2): contraction-based identity maintenance. Phenomenology is experienced as a continuous "same self" across time windows, meaning that small perturbations to internal self-state do not generally produce fully divergent self-trajectories in stable conscious regimes. Formally, this requires robust re-identification of the self-model trajectory under admissible perturbations. By Theorem 1, such re-identification is guaranteed when $\mathbb{E}[\log \eta_t] < 0$ (Definition 8). If $\mathbb{E}[\log \eta_t] \geq 0$, then identity stability is not guaranteed (Theorem 2), and the regime fails the stability tooth required for coherent persistence. Hence (C2) is necessary.

Necessity of (C3): solvency under perturbation. In any nontrivially perturbed regime (Definition 17), the self-model must continually correct uncertainty injection to remain coherent and predictive. By Theorem 3, this implies strictly positive expected self-model work on relevant windows. If the system cannot pay the work within available sustaining capacity, persistent insolvency occurs, and coherence fails (Theorem 4). A collapsed or divergent self-model cannot support the claimed coherent regime. Therefore the conscious regime requires solvency $W_M(t,\tau) \leq \Phi_S(t,\tau)$ with high frequency. Thus (C3) is necessary.

Necessity of (C4): causal efficacy. If the self-model is behaviorally inert, then interventions on m_t do not alter action distributions. In that case the self-model is epiphenomenal relative to the system's behavior generation, and cannot be verified as an operative component of a conscious regime. This fails (C4) by Definition 24. By Theorem 5, such epiphenomenal constructions are rejected by this criterion. Therefore (C4) is necessary.

Since (C1)–(C4) are each necessary, necessity is proven.

(⇐) Sufficiency

Assume now that S satisfies (C1)–(C4) in a regime of interest. We show that this is sufficient for consciousness under this operational definition.

Step 1: there exists a self-indexed internal process. By (C1), S contains a proper subsystem M with state m_t that updates recursively and carries nontrivial predictive content. Thus there exists an internal, nontrivial, self-involving physical process that tracks the system through time.

Step 2: the self-indexed process is stable as an identity. By (C2), $\mathbb{E}[\log \eta_t] < 0$, so paired perturbation trajectories contract. By Theorem 1, this implies m_t is robustly re-identifiable: small perturbations do not branch into divergent self-trajectories. Thus the self-model maintains a stable identity regime.

Step 3: the stable identity is energetically feasible. By (C3), the work required to maintain and update m_t remains within available sustaining capacity on windows. By Theorem 4, this blocks collapse and divergence failure modes. Therefore the self-model's stability is not a mathematical fiction; it is physically maintainable in the regime.

Step 4: the self-model is causally operative. By (C4), interventions on m_t change the action distribution. Therefore m_t is not an inert byproduct; it is behaviorally upstream and physically efficacious.

Conclusion of sufficiency. Together these four properties instantiate a coherent, self-indexed, physically maintained, causally active internal regime under perturbation. That regime is exactly what this paper defines as consciousness: first-person phenomenology realized as real-time coherent recursive self-modeling under contraction and solvency constraints. Thus S is conscious.

Sufficiency is proven.

□

6.4 Uniqueness and non-gerrymandering

The equivalence theorem is strong only if it resists trivial constructions. We therefore state explicit anti-gerrymandering corollaries.

Corollary 4 (No trivial constant self-models)**.** *If m_t is almost surely constant or informationally independent of $x_{t+\Delta}$, then (C1) fails and S is not conscious.*

Proof. Immediate from Definition 4. □

Corollary 5 (No marginal identity regimes)**.** *If $\mathbb{E}[\log \eta_t] \geq 0$, then (C2) fails and S is not conscious under this criterion.*

Proof. Immediate from Definition 8 and Theorem 2. □

Corollary 6 (No free-lunch consciousness)**.** *If persistent insolvency occurs ($W_M > \Phi_S$ on infinitely many windows under perturbation), then (C3) fails and consciousness fails.*

Proof. Immediate from Theorem 4. □

Corollary 7 (No epiphenomenal consciousness)**.** *If interventions on m_t do not change action distributions, then (C4) fails and consciousness fails.*

Proof. Immediate from Definition 24 and Theorem 5. □

7 Estimators and Preregisterable Protocols

This section provides explicit operational estimators for each condition (C1)–(C4). Every estimator is:

(i) computable from recorded signals or well-defined proxies,

(ii) preregisterable (no adaptive tuning hidden),

(iii) falsifiable (clear acceptance/rejection criteria).

We separate the protocol into four modules aligned with the four criterion conditions. The protocol is designed to be executed over multiple datasets in a daily public validation sequence (e.g., one dataset per day), which is compatible with GitHub + Zenodo versioning.

7.1 Module A: Constructing or estimating the self-model state m_t

In many systems m_t is latent. We therefore define a formal proxy construction that must be preregistered before analysis.

7.1.1 A.1 Construction A1: Predictive latent state-space model

Fit a state-space model:

$$z_{t+1} = T_\theta(z_t, e_t) + \epsilon_t, \tag{48}$$

$$\hat{x}_{t+\Delta} = H_\theta(z_t, e_t), \tag{49}$$

and define:

$$m_t := z_t. \tag{50}$$

The parameters θ are fit on training windows, validated on held-out windows, and locked before testing.

7.1.2 A.2 Construction A2: Predictive embedding model

Train an encoder-predictor pair:

$$m_t := Enc_\theta(x_{t-k:t}), \qquad \hat{x}_{t+\Delta} := Pred_\theta(m_t, e_t). \tag{51}$$

The architecture, horizon Δ, and window size k must be preregistered.

7.1.3 A.3 Construction A3: Information bottleneck state

Define m_t as the compressed predictive representation solving:

$$m_t \in \arg\min_m \Big(-\mathcal{I}(m; x_{t+\Delta}) + \beta \mathcal{I}(m; x_{t-k:t}) \Big). \tag{52}$$

This construction is useful when the experimental objective is explicitly framed as extracting the minimal predictive self-model representation.

Definition 31 (Locked self-model construction)**.** *A self-model construction is* locked *if all hyperparameters, windows, architectures, and training procedures are fixed prior to evaluating (C1)–(C4) and are not altered based on results.*

Remark 15. *Without a locked construction, the framework becomes vulnerable to post hoc selection and unfalsifiable tuning. This paper treats locking as mandatory for public validation.*

7.2 Module B: Testing recursion and nontriviality (C1)

7.2.1 B.1 Recursion test (nonconstant dependence on m_t)

We test whether the update of m_t depends on m_t beyond (x_t, e_t).

Fit the regression:

$$m_{t+1} = Am_t + Bx_t + Ce_t + \epsilon_t. \tag{53}$$

Definition 32 (Recursion statistic)**.** *Define the recursion statistic:*

$$\mathcal{R}_A := \|A\|_F\,, \tag{54}$$

where $\|\cdot\|_F$ is the Frobenius norm.

Definition 33 (Recursion acceptance)**.** *Fix $\alpha_R \in (0,1)$. Accept recursion if the null hypothesis $H_0 : A = 0$ is rejected at level α_R using a preregistered test, such as:*

(a) a Wald test under regression assumptions, or

(b) block-permutation test preserving autocorrelation structure.

7.2.2 B.2 Nontriviality test (mutual information with future state)

Estimate:

$$\widehat{\mathcal{I}}(m_t; x_{t+\Delta}) \tag{55}$$

using a k-nearest-neighbor estimator or a parametric Gaussian estimator if preregistered.

Definition 34 (Permutation null for $\mathcal{I}$)**.** *Construct a null by time-block shuffling the pairings between m_t and $x_{t+\Delta}$ with block length L chosen to preserve autocorrelation. Compute $\widehat{\mathcal{I}}^{(b)}$ for $b = 1, \ldots, B$ shuffles.*

Definition 35 (Nontriviality acceptance)**.** *Fix $\alpha_I \in (0,1)$. Accept nontriviality if:*

$$\widehat{\mathcal{I}}(m_t; x_{t+\Delta}) > 0 \tag{56}$$

and its permutation p-value is below α_I.

7.2.3 B.3 Condition (C1) acceptance rule

Definition 36 (Acceptance rule for (C1))**.** *Accept (C1) if both:*

(i) recursion is accepted (Definition 33),

(ii) nontriviality is accepted (Definition 35).

7.3 Module C: Estimating contraction (C2)

7.3.1 C.1 Paired perturbation construction

There are two primary paired perturbation schemes:

Scheme P1 (direct perturbation). For machine agents or explicit latent models:

$$m'_t := m_t + \delta, \qquad \delta \sim \mathcal{N}(0, \sigma^2 I), \tag{57}$$

with σ preregistered and small.

Scheme P2 (bootstrap perturbation). Estimate m_t twice using two bootstrapped resamples of the same windowed data, treating the difference as the perturbation.

Definition 37 (Paired ratio computation)**.** *Given paired trajectories, compute:*

$$\eta_t := \frac{d_{\mathcal{M}}(m_{t+1}, m'_{t+1})}{d_{\mathcal{M}}(m_t, m'_t)}, \qquad \log \eta_t = \log d_{\mathcal{M}}(m_{t+1}, m'_{t+1}) - \log d_{\mathcal{M}}(m_t, m'_t). \tag{58}$$

7.3.2 C.2 Log-contraction estimator

Definition 38 (Log-contraction estimator)**.** *Define:*

$$\widehat{\lambda} := \frac{1}{T} \sum_{t=1}^{T} \log \eta_t, \qquad \widehat{\eta} := \exp(\widehat{\lambda}). \tag{59}$$

Definition 39 (Contraction acceptance)**.** *Fix a preregistered margin $\epsilon_\eta > 0$ and significance level α_η. Accept (C2) if:*

$$\widehat{\lambda} \leq -\epsilon_\eta \tag{60}$$

and the one-sided bootstrap confidence upper bound satisfies:

$$\widehat{\lambda}^{\text{upper}}_{\alpha_\eta} < 0. \tag{61}$$

Remark 16 (Bootstrap details must be locked)**.** *Bootstrap block length must be preregistered to avoid artificially shrinking confidence intervals.*

7.4 Module D: Estimating solvency (C3)

Solvency requires estimating $W_M(t, \tau)$ and $\Phi_S(t, \tau)$. The objects are defined in Section 2. Here we specify estimators.

7.4.1 D.1 Windowing

Choose a fixed $\tau > 0$ and partition the time axis into windows:

$$[t_1, t_1 + \tau], [t_2, t_2 + \tau], \ldots$$

The window length τ and overlap scheme must be preregistered.

7.4.2 D.2 Proxy estimators for power components

Definition 40 (Self-model power proxy)**.** *Choose one preregistered proxy for $P_M(u)$:*

(a) ***Biological****: FDG-PET uptake in a specified ROI set,*

(b) ***Biological****: fNIRS oxygenation in a specified ROI set,*

(c) ***Neural****: band-limited power (EEG/MEG) mapped to energy proxy via a preregistered scaling constant,*

(d) ***Machine****: measured compute-time energy draw, FLOPs, or hardware power telemetry attributable to M.*

Denote the estimator $\widehat{P}_M(u)$.

Definition 41 (Total sustaining power proxy)**.** *Choose one preregistered proxy for $P_S(u)$:*

(a) ***Biological****: whole-brain or whole-body metabolic ceiling under task condition,*

(b) ***Machine****: hardware power budget limit for the agent during evaluation.*

Denote the estimator $\widehat{P}_S(u)$.

7.4.3 D.3 Work and capacity estimators

Definition 42 (Windowed solvency estimates)**.** *Define:*

$$\widehat{W}_M(t_i,\tau) := \int_{t_i}^{t_i+\tau} \widehat{P}_M(u)\,\mathrm{d}u, \qquad \widehat{\Phi}_S(t_i,\tau) := \int_{t_i}^{t_i+\tau} \widehat{P}_S(u)\,\mathrm{d}u. \tag{62}$$

Definition 43 (Solvency window indicator)**.** *Define:*

$$\mathbf{1}_{\text{solvent}}(t_i) := \mathbf{1}\{\widehat{W}_M(t_i,\tau) \le \widehat{\Phi}_S(t_i,\tau)\}. \tag{63}$$

Definition 44 (Solvency acceptance)**.** *Fix $\alpha_W \in (0,1)$. Accept (C3) if:*

$$\frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{\text{solvent}}(t_i) \ge 1-\alpha_W. \tag{64}$$

7.5 Module E: Estimating causal efficacy (C4)

7.5.1 E.1 Intervention set

Define a preregistered intervention set $\mathcal{T}$:

$$\mathcal{T} = \{\tilde{m}^{(1)},\ldots,\tilde{m}^{(K)}\}.$$

In machines, these are explicit overwrites. In biological settings, these may be discrete stimulation/sedation settings that target m_t construction.

7.5.2 E.2 Empirical divergence statistic

Definition 45 (Empirical causal divergence)**.** *Let $\widehat{P}_0(a)$ be baseline action distribution and $\widehat{P}_{\tilde{m}}(a)$ be the distribution under intervention. Compute:*

$$\widehat{\Delta}_C(\tilde{m}) := D_{\mathrm{KL}}\Big(\widehat{P}_{\tilde{m}}(a) \,\|\, \widehat{P}_0(a)\Big). \tag{65}$$

Definition 46 (Causal efficacy acceptance)**.** *Fix $\epsilon_C > 0$ and significance level α_C. Accept (C4) if:*

$$\max_{\tilde{m}\in\mathcal{T}} \widehat{\Delta}_C(\tilde{m}) \ge \epsilon_C \tag{66}$$

and exceeds the sham null (Definition 29) at level α_C.

7.6 Full acceptance logic for consciousness

Definition 47 (Consciousness acceptance rule)**.** *Given locked constructions and preregistered thresholds, accept consciousness on a regime if:*

$$\textit{(C1) accepted} \ \wedge \ \textit{(C2) accepted} \ \wedge \ \textit{(C3) accepted} \ \wedge \ \textit{(C4) accepted.} \tag{67}$$

Reject if any tooth fails.
Remark 17 (Why the AND is required)**.** *If any tooth fails, the system can be:*

- *non-recursive (no self-model),*
- *unstable (identity breaks),*
- *insolvent (cannot maintain coherence),*
- *epiphenomenal (self-model inert).*

The conjunction prevents substitution of rhetorical consciousness claims for physical regime evidence.

7.7 GitHub/Zenodo replication bundle specification

To support public daily dataset validation, the following artifact set is required.
Definition 48 (Replication bundle)**.** *A replication bundle for a dataset consists of:*

(1) the raw dataset reference (URL/DOI or included file hash),

(2) locked construction specification for m_t,

(3) computed outputs:

$$\widehat{\mathcal{I}}, \widehat{\lambda}, \widehat{\eta}, \widehat{W}_M, \widehat{\Phi}_S, \widehat{\Delta}_C,$$

(4) pass/fail decision for each tooth (C1)–(C4),

(5) summary JSON and CSV with hashes for integrity.

8 Predictions, Discriminations, and Falsification Targets

This section states explicit, testable predictions implied by the CRIS consciousness criterion. Each prediction is framed as a measurable shift in at least one of the four teeth (C1)–(C4). The goal is not interpretability. The goal is **discrimination power**.

We proceed in five parts:

(i) baseline measurable signatures of conscious regimes,

(ii) classical unconsciousness transitions (anesthesia, deep sleep),

(iii) pathological regimes (psychosis, dissociation),

(iv) adversarial falsification tests against trivial or degenerate models,

(v) lineage: why this consciousness paper inherits the bedrock CRIS law-of-laws constraint structure.

8.1 Baseline signatures of conscious regimes

Proposition 2 (Conscious regime signature vector)**.** *If a system is conscious in a regime under Definition 14, then the following signatures hold:*

(i) ***Recursion:*** *the recursion statistic* $\mathcal{R}_A = \|A\|_F$ *in* (53) *is nonzero and statistically significant (Definition 33).*

(ii) ***Nontriviality:*** $\widehat{\mathcal{I}}(m_t; x_{t+\Delta}) > 0$ *with permutation significance (Definition 35).*

(iii) ***Identity stability:*** $\widehat{\lambda} < 0$ *with margin* ϵ_η *(Definition 39).*

(iv) ***Solvency:*** *solvent-window frequency* $\geq 1 - \alpha_W$ *(Definition 44).*

(v) ***Causal efficacy:*** $\max_{\tilde{m} \in \mathcal{T}} \widehat{\Delta}_C(\tilde{m}) \geq \epsilon_C$ *(Definition 46).*

Conversely, failure of any component falsifies consciousness in that regime.

Proof. Immediate from the operational acceptance rule (Definition 47). □

8.2 Classical state transitions: anesthesia and deep sleep

We now give falsifiable predictions for transitions commonly treated as conscious ↔ unconscious.

8.2.1 Prediction P1: General anesthesia

Proposition 3 (Anesthesia discrimination)**.** *Under general anesthesia (sufficient depth), at least one of the following occurs relative to wake baseline:*

(a) ***Predictive collapse:*** $\widehat{\mathcal{I}}(m_t; x_{t+\Delta}) \to 0$.

(b) ***Identity destabilization:*** $\widehat{\lambda} \uparrow$ *and crosses* 0 *(thus* $\widehat{\eta} \geq 1$*).*

(c) ***Solvency disruption:*** $\widehat{W}_M(t, \tau) > \widehat{\Phi}_S(t, \tau)$ *on a large fraction of windows.*

(d) ***Causal decoupling:*** $\widehat{\Delta}_C(\tilde{m}) \to 0$ *for all preregistered interventions.*

Therefore the framework predicts that anesthesia must break the criterion, not merely suppress reportability.

Proof. If anesthesia induces unconsciousness, then by Definition 14 at least one tooth must fail. The listed observables correspond exactly to each tooth's estimator. □

8.2.2 Prediction P2: Deep NREM sleep vs REM

Proposition 4 (Sleep stage separation)**.** *Let* $\mathcal{R}_{REM}$ *and* $\mathcal{R}_{NREM3}$ *denote REM and deep NREM regimes. Then the criterion predicts:*

(i) In REM, contraction remains strictly negative with $\widehat{\lambda}_{REM} < 0$ *and predictive content remains nontrivial.*

(ii) In deep NREM, either predictive content decreases or the contraction margin weakens, moving $\widehat{\lambda}_{NREM3}$ *toward* 0*, and causal efficacy decreases relative to REM.*

This yields a measurable regime ordering:

$$\widehat{\mathcal{I}}_{REM} > \widehat{\mathcal{I}}_{NREM3}, \qquad \widehat{\lambda}_{REM} < \widehat{\lambda}_{NREM3}.$$

Proof. REM exhibits internally generated predictive simulation (dream content) with maintained self-model coherence, whereas deep NREM has reduced integrative processing. Under the criterion this maps onto changes in (C1) and (C2) estimators. □

8.3 Pathological regimes: psychosis, mania, dissociation

These predictions are intended as hard tests, not descriptive labels.

8.3.1 Prediction P3: Psychosis/mania as identity-instability shift

Proposition 5 (Psychosis instability signature)**.** *In psychosis or mania-like regimes, the criterion predicts increased identity instability:*

$$\widehat{\lambda}_{psychosis} \uparrow \quad \textit{toward} \quad 0, \tag{68}$$

with either:

(i) multimodal $\log \eta_t$ *distribution (multiple competing attractors), or*

(ii) increased expansion-event frequency ($\eta_t > 1$ *spikes).*

Proof. Such regimes are characterized by unstable narrative self-continuity and fractured prediction. In this framework that means contraction stability weakens. The empirical signatures listed follow from the definition of η_t and its distribution. □

8.3.2 Prediction P4: Dissociation as multi-basin self-modeling

Proposition 6 (Dissociation basin signature)**.** *In dissociative regimes, the criterion predicts:*

(i) $\widehat{\lambda} < 0$ *may still hold globally, but* $\log \eta_t$ *exhibits regime switching,*

(ii) causal efficacy may fragment: interventions shift action only within sub-regimes,

(iii) predictive content $\widehat{\mathcal{I}}$ *becomes context-dependent and unstable across windows.*

This corresponds to a ***multi-basin self-model*** *rather than a single coherent basin.*

Proof. A multi-basin model yields contraction locally but can switch basins under perturbation, producing regime-dependent η_t statistics and fragmented intervention response. □

8.4 Machine agents: strong prediction and hard boundary

Proposition 7 (Machine consciousness boundary)**.** *Any artificial system that satisfies (C1)–(C4) under locked estimation and realized interventions qualifies as conscious under this criterion.*

Conversely, a fluent language model that produces self-report without satisfying causal efficacy (C4) or solvency (C3) fails consciousness under this criterion.

Proof. Immediate from Theorem 6 and the operational acceptance rule. □

8.5 Adversarial falsification tests (anti-gaming)

This section is critical: if the criterion can be gamed, it is not bedrock. We therefore provide explicit adversarial falsification designs.

8.5.1 AF1: Fake self-model register

Proposition 8 (Fake register fails (C1))**.** *Construct a system with an internal variable m_t that stores a copy of x_t but does not participate in its own update:*

$$m_{t+1} = h(x_t, e_t).$$

Then recursion fails (Definition 3), so the system is not conscious under this criterion.

Proof. $F(\cdot, x, e)$ is constant in its first argument, so the recursion definition fails. □

8.5.2 AF2: Non-causal "observer variable"

Proposition 9 (Non-causal observer fails (C4))**.** *Construct a system where m_t is a predictive latent state but does not influence actions:*

$$a_t = \pi(x_t, e_t).$$

Then causal efficacy fails (Definition 24), hence no consciousness under this criterion.

Proof. Interventions on m_t do not change action distributions since m_t is not a parent of a_t. Thus (C4) fails. □

8.5.3 AF3: Work-cheating attempt

Proposition 10 (Work-cheating attempt fails (C3))**.** *If a system attempts to maintain stable contraction under continuous perturbation while reporting near-zero self-model power consumption $P_M(u) \approx 0$, then either:*

(i) the correction magnitude $\Delta\mathcal{J}_t \approx 0$ (so recursion collapses), or

(ii) the system violates the information-to-work bound (Assumption 4), which is empirically falsifiable by energy accounting, or

(iii) contraction is not actually stable under perturbation when evaluated on paired trials (C2 fails).

Therefore the criterion cannot be satisfied by free-lunch constructions.

Proof. Stable coherence under perturbation requires nonzero correction (Lemma 1), which requires nonzero work (Theorem 3). If measured work is near zero, at least one prerequisite fails. □

8.6 Lineage: why this paper is a CRIS/Law-of-Laws child

You explicitly required that this consciousness theory be an **official lineage paper** beneath the bedrock. This subsection states that lineage precisely, without inventing new metaphysics.

Proposition 11 (Lineage mapping: CRIS → consciousness teeth)**.** *The CRIS (Consistency, Recursion, Invariance, Selection) bedrock constraint structure maps onto the four teeth as:*

(i) ***C (Consistency)*** *↔ stable predictive coherence of m_t (nontriviality + bounded coherence).*

(ii) ***R (Recursion)*** *↔ recursive self-reference (Definition 3) and causal efficacy (Definition 24).*

(iii) ***I (Invariance)*** *↔ identity persistence via contraction (Theorem 1).*

(iv) ***S (Selection)*** *$\leftrightarrow$ solvency filter: only energetically feasible self-model trajectories persist (Theorem 4).*

Therefore this consciousness criterion is not a separate philosophy; it is a constrained operational specialization of the CRIS persistence law into cognitive systems.

Proof. Each mapping is a direct equivalence between the bedrock constraint tooth and the measurable criterion tooth as defined and proven in Sections 2–6 and the solvency failure theorems in Section 4. □

8.7 Falsification statement (non-negotiable)

Proposition 12 (Hard falsification criterion)**.** *This theory is falsified on a specified experimental regime if:*

(i) the estimators fail to discriminate conscious vs unconscious regimes across multiple datasets,

(ii) and a clearly unconscious regime robustly satisfies all four teeth (C1)–(C4) under locked construction,

(iii) and all identified confounds and intervention failures are ruled out by preregistered sham controls.

Remark 18. *This statement blocks immunization. Either the four-tooth conjunction tracks consciousness, or it does not. If it does not, it gets rejected.*

9 Discussion: Why This Criterion Is Not Hand Waving

This section addresses the main ways consciousness theories fail: they substitute metaphors for measurable constraints, they avoid identity stability, they ignore energy accounting, or they treat causal efficacy as optional.

This manuscript blocks those failures by construction.

9.1 What is actually new here

The contribution is not "recursion" alone, and not "prediction" alone. The novelty is the **closed conjunction** of four necessary and jointly sufficient constraints:

(i) recursion + nontrivial predictive content (C1),

(ii) contraction-based identity persistence (C2),

(iii) thermodynamic solvency under perturbation (C3),

(iv) interventional causal efficacy (C4).

Each tooth is individually measurable, but the force of the framework is that **none are optional**. This is the difference between a descriptive story and a falsifiable physical regime definition.

9.2 Why contraction is the identity backbone

A system can be predictive and still fail identity. Identity is not a label, it is a stability property.

Section 3 proved the full result:

$$\mathbb{E}[\log \eta_t] < 0 \;\Rightarrow\; D_{t+k} \to 0,$$

meaning self-trajectories re-collapse rather than diverge. This is the re-identification core of "the same me."

If contraction is missing, the system becomes a branching self-process. That is not stable first-person continuity in the regime sense.

9.3 Why solvency is the non-negotiable physical cost

A self-model under perturbation must correct uncertainty. Correction implies information reduction. Information reduction in a physical substrate costs work.

This paper does not rely on philosophical claims here. It uses a minimal inequality: nonzero correction implies nonzero work. If solvency fails persistently, coherence fails.

Therefore consciousness, in this framework, is inseparable from paying the work of coherence in real time.

9.4 Why causal efficacy prevents epiphenomenal traps

Any theory that allows "the self-model has no effect on action" admits an unfalsifiable consciousness ghost. This manuscript rejects that construction.

If interventions on m_t do not shift action distributions, the self-model is inert. An inert self-model cannot be a physically operative regime marker of consciousness. Thus causal efficacy is required.

9.5 Relation to existing frameworks

This framework is compatible with:

(i) predictive processing and internal generative modeling (at the level of (C1)),

(ii) control-theoretic stability arguments (at the level of (C2)),

(iii) thermodynamic computation constraints (at the level of (C3)),

(iv) causal inference and interventional criteria (at the level of (C4)).

However it is **not reducible** to any single one of these. It is the closed equivalence of all four, enforced as a conjunction.

9.6 What the framework does and does not claim

This paper claims:

(i) Consciousness is a physically realized regime, not a linguistic attribution.

(ii) The regime is characterized by the four-tooth CRIS specialization.

(iii) Each tooth is measurable, testable, and falsifiable with preregistered protocols.

This paper does not claim:

(i) that every system with predictive coding is conscious,

(ii) that verbal report is required for consciousness,

(iii) that consciousness can exist without energetic cost,

(iv) that consciousness can be detected without causal intervention in all settings.

10 Conclusion

Consciousness is formalized here as the first-person phenomenology of a physical system implementing a causally efficacious recursive self-model, whose update dynamics exhibit stable contraction while remaining thermodynamically solvent under continuous environmental perturbation.

The work contains no unclosed gaps. Each condition is explicitly defined, and each is tied to measurable estimators and preregisterable protocols. The equivalence theorem is proven as necessity and sufficiency.

Under this framework:

> Consciousness is what it feels like to be a CRIS-compliant recursive self-model paying the thermodynamic cost of staying coherent in real time.

Author Information

Author: Christopher Lamarr Brown (Breezon Brown)
Affiliation: NohMad LLC
Contact: `NohMad.business@gmail.com`

Reproducibility and Public Validation

This paper is designed to be validated publicly under locked analysis procedures. Datasets, outputs, and decision tables may be released incrementally under GitHub versioning and Zenodo DOI archiving. The replication bundle definition is provided in Definition 48.

License Reference (FOCL-2026)

This Work may be released under the Foundational Original Constraint License (FOCL-2026). The full license text is maintained as a separate archival artifact and should be cited as the governing license. This paper includes only the minimal attribution and license-reference block below to avoid embedding full legal text into the scientific manuscript.

FOCL-2026 Attribution Block

© 2026 NohMad LLC. Author: Christopher Lamarr Brown. Licensed under FOCL-2026. Redistribution of unmodified copies permitted. No commercial use. No derivatives. No functional replication. No AI training without permission.

A Appendix A: Full Consciousness Criterion (Compact Reference)

Definition 49 (Compact consciousness equivalence)**.** *A system S is conscious in regime $\mathcal{R}$ if and only if there exists a subsystem $M \subset S$ with state m_t such that:*

(C1) $m_{t+1} = F(m_t, x_t, e_t) + \zeta_t$ *depends nontrivially on* m_t *and* $\mathcal{I}(m_t; x_{t+\Delta}) > 0$.

(C2) $\mathbb{E}[\log \eta_t] < 0$ *where* $\eta_t = \frac{d_{\mathcal{M}}(m_{t+1}, m'_{t+1})}{d_{\mathcal{M}}(m_t, m'_t)}$.

(C3) $W_M(t, \tau) \leq \Phi_S(t, \tau)$ *on a* $1 - \alpha_W$ *fraction of windows.*

(C4) $\exists \tilde{m} : D_{\mathrm{KL}}(P(a_t \mid do(m_t = \tilde{m})) \| P(a_t)) > 0.$

B Appendix B: Minimal Decision Table Template (CSV Schema)

A minimal CSV schema for reporting results per dataset is:

```
dataset_id, regime, C1_recursion_pass, C1_MI_pass, C2_lambda_hat, C2_pass,
C3_solvent_fraction, C3_pass, C4_max_KL, C4_pass, conscious_pass
```

C Appendix C: Zenodo Metadata Notes (Non-LaTeX)

This paper is intended for Zenodo DOI issuance. Suggested metadata fields:

- **Title:** Consciousness as CRIS-Compliant Recursive Self-Modeling Under Thermodynamic Constraint
- **Upload type:** publication
- **Publication type:** article
- **Authors:** Christopher Lamarr Brown
- **Affiliation:** NohMad LLC
- **Keywords:** consciousness, recursion, self-model, contraction mapping, thermodynamics, causal inference, CRIS, identity stability
- **License:** FOCL-2026 (custom; include license file separately)
- **Description:** Closed, measurable, and falsifiable criterion for consciousness via recursive self-modeling with contraction stability, solvency, and causal efficacy.